DataSight PRO

Streamlining Data Analysis & ML Preparation

An Interactive Tool for Exploration, Visualization, and Preprocessing

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INEW 2332 - Comprehensive Software Project

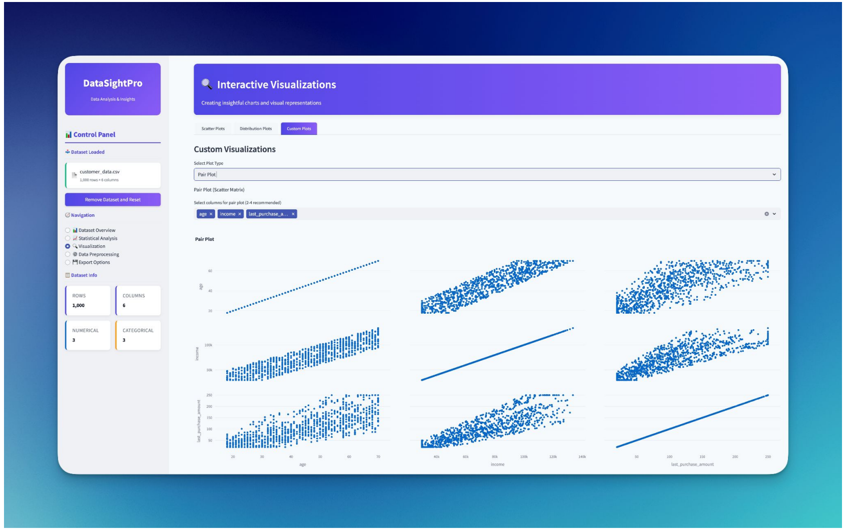
The Challenge & The Solution

Challenge: Data analysis and preparation for Machine Learning can be complex, time-consuming, and require significant coding expertise.

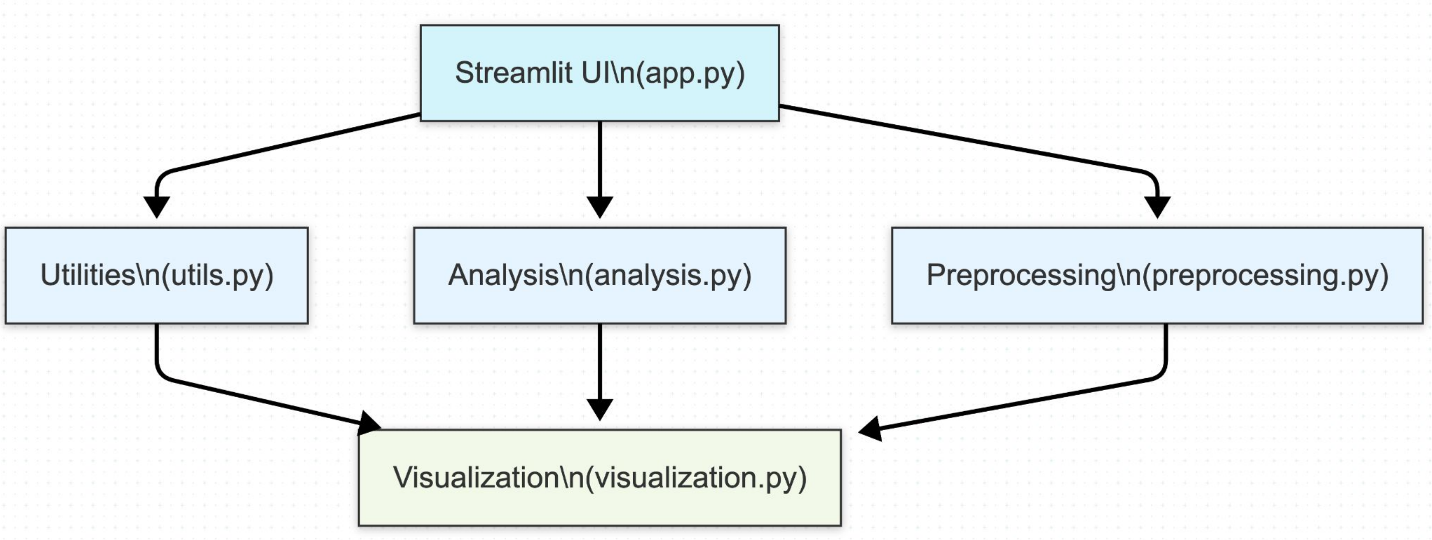
Solution: DataSight PRO - An interactive web application built with Python & Streamlit.

Goal: To streamline the data analysis workflow by providing an intuitive interface for exploration, statistical analysis, visualization, and preprocessing – making it accessible without extensive programming.

Target Audience: Data Scientists, Analysts, ML Engineers, Researchers, Students, Anyone working with data.

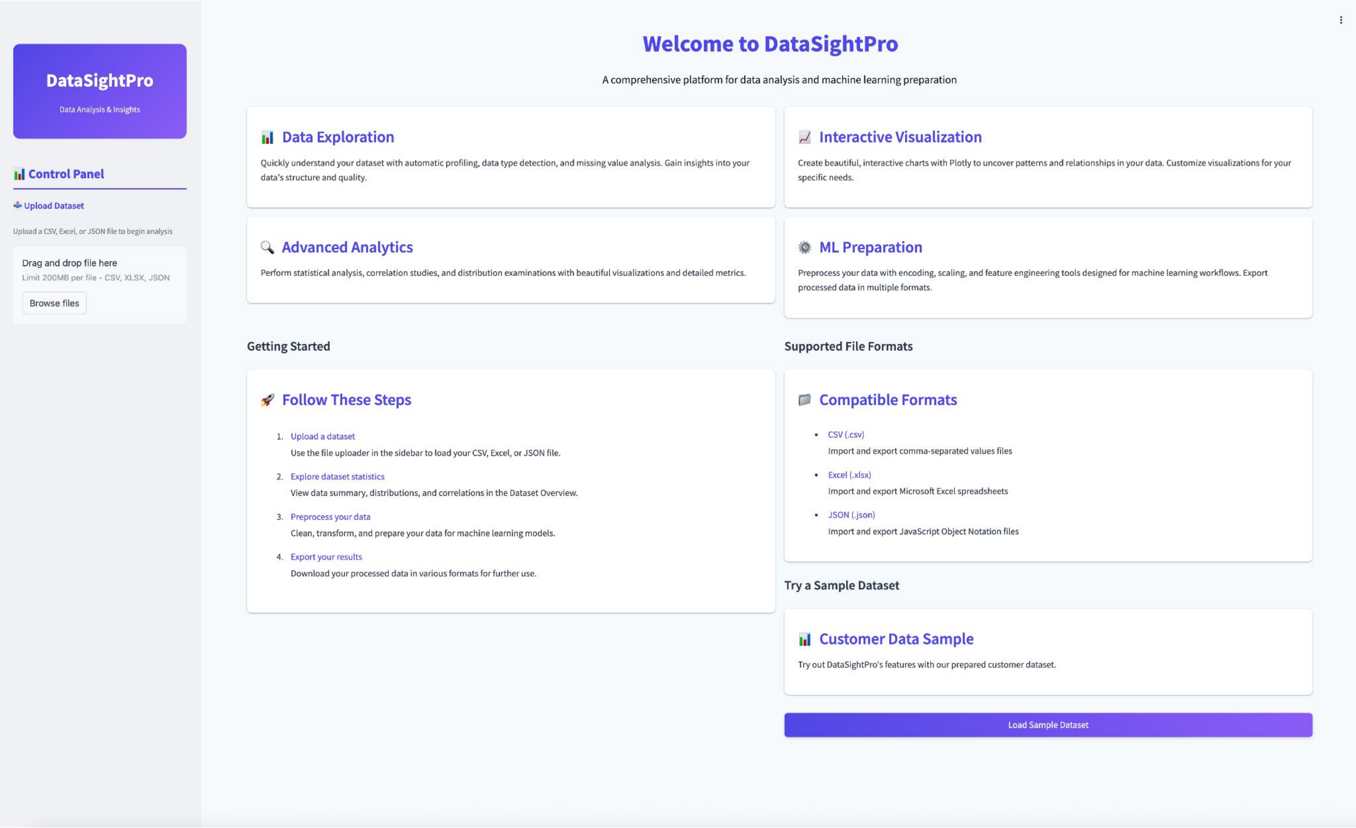


Architecture Overview



The Streamlit UI layer interacts with distinct backend modules -Utilities, Analysis, Preprocessing, and Visualization. The main application coordinates data flow between these modules..

Intuitive UI

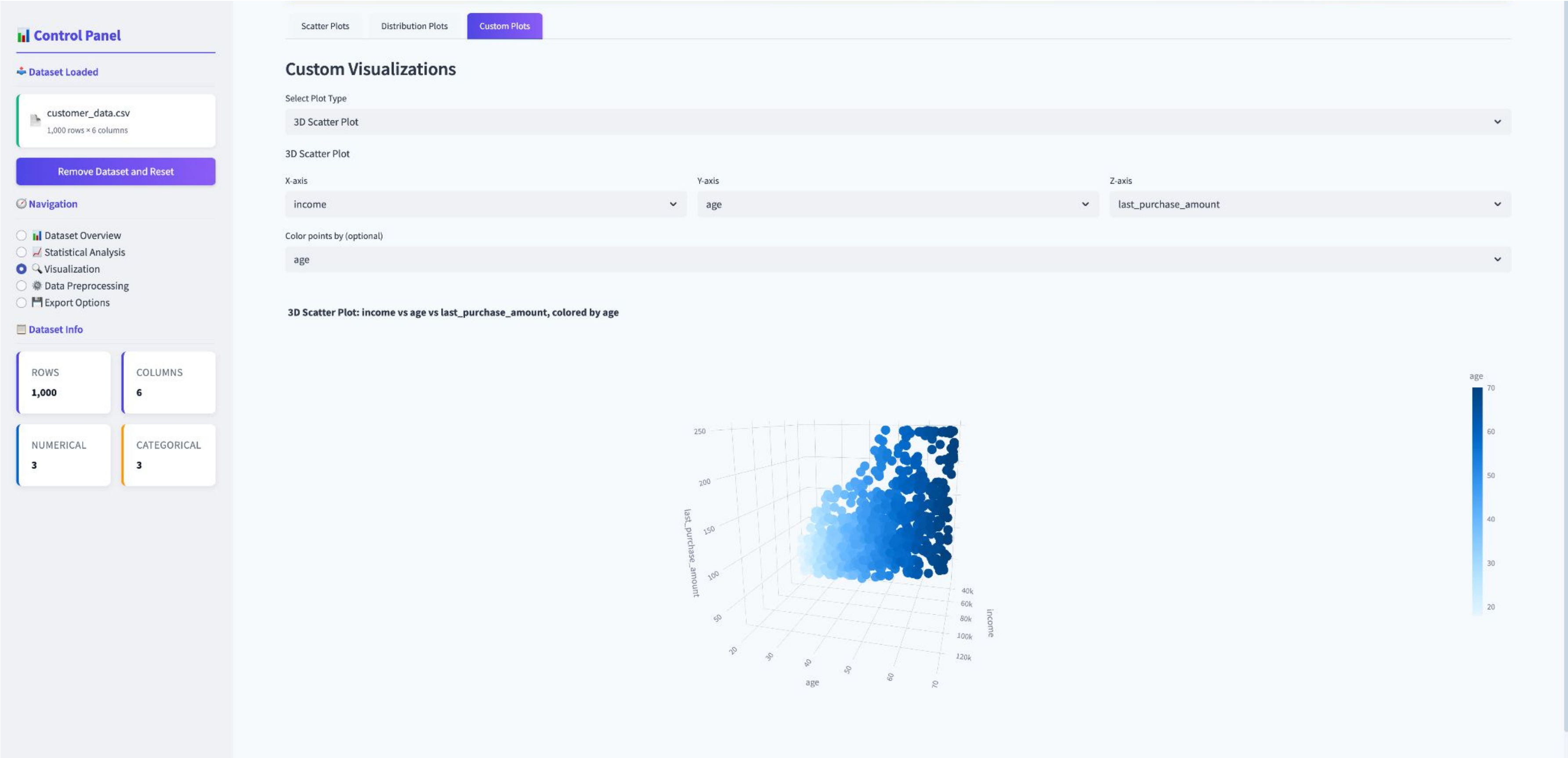


DataSight PRO Workflow

🚀 Follow These Steps

1. Upload a dataset
   Use the file uploader in the sidebar to load your CSV, Excel, or JSON file.
2. Explore dataset statistics
   View data summary, distributions, and correlations in the Dataset Overview.
3. Preprocess your data
   Clean, transform, and prepare your data for machine learning models.
4. Export your results
   Download your processed data in various formats for further use.

Uncover Insights Visually



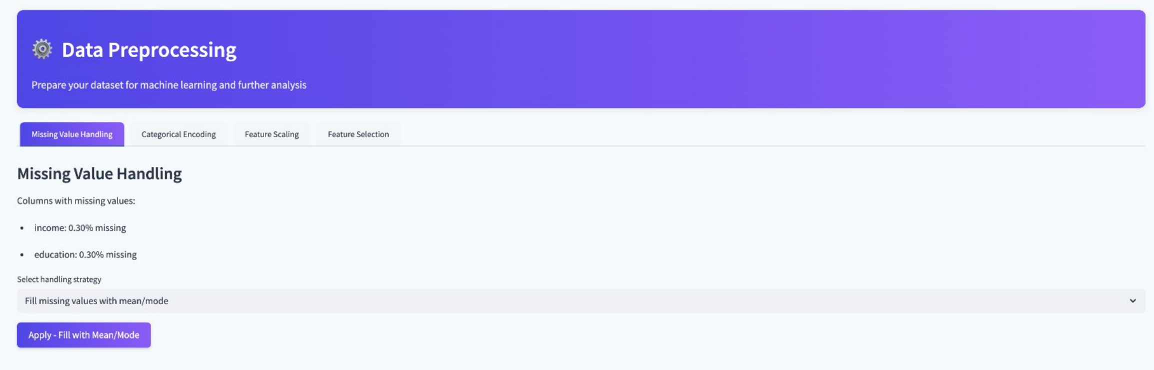
Prepare Data for Machine Learning

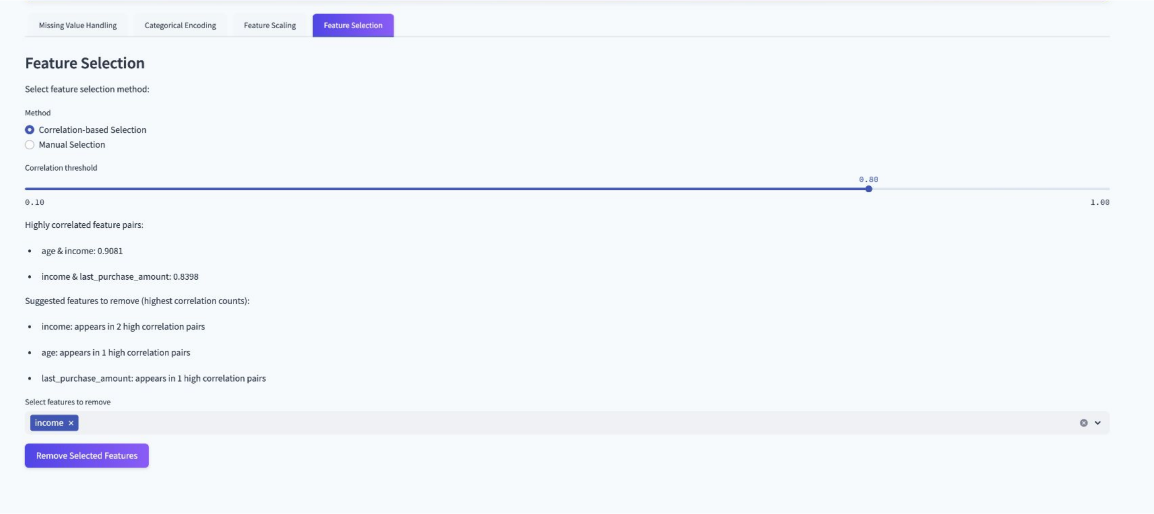
Handle Missing Values: Impute using Mean, Median, Mode, or Constant Value

Encode Categorical Data: Convert text to numbers using One-Hot or Label Encoding

Normalize Data: Normalize data using MinMax, Z-Score, or Robust Scaling (handles outliers)

Feature Selection: Select variables manually or by correlation





Instant Insights, Endless Evolution

Core Benefits:

- **Speed:** Saves time and delivers instant results
- **Simplicity:** Reduces complexity of data analysis via code-minimal interface
- **Accessibility:** Empowers users with diverse skill levels

Future Enhancements:

- AutoML Integration
- Advanced Visualizations
- Direct Database Connectors
- Guided Workflows & Customizable Dashboards

DataSight PRO

Empowering Insights for Everyone

Enjoy your Data!

GitHub Repo

Project Demo